Kunal Pai

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Education

Ph.D., Computer Science, University of California, Los Angeles — Expected: June 2032
M.S., Computer Science, University of California, Davis (GPA: **4.0/4.0**) — June 2026
B.S., Computer Science & Engineering, University of California, Davis (GPA: **3.8/4.0**), *with Honors* — June 2023

Relevant Skills

Languages: Python, C++, C, JavaScript, Java, Rust
ML/AI: TensorFlow, PyTorch, scikit-learn, LLMs, Prompt Engineering, Ollama, Hugging Face, Multi-agent Systems
Web/Data: React, Next.js, Django, Flask, MongoDB, pandas, NumPy, Matplotlib
Tools: Git, Docker, Unix/Linux, gem5, Jupyter, LLVM, Clang

Work Experience

Graduate Student Researcher, *DavSec Lab @ UC Davis* — April 2025 – Present
Research Project — *Python, LLMs, C++, Rust, Compiler Analysis*
- Built automated C-to-Rust transpilation pipeline using LLMs with 5 prompt variations, benchmarking state-of-the-art models across 746 programs and achieving 70.2% functional accuracy
- Engineered a multi-variant performance-aware transpilation framework that explores alternative Rust implementations in an evolutionary agent setting, yielding up to 3× runtime reduction across benchmarks
- Developed cross-layer analysis framework combining compiler metrics with hardware performance counters to expose semantic and optimization gaps in LLM-translated, syntax-directed and manually-written Rust translations

Graduate Student Researcher, *DECAL Lab @ UC Davis* — September 2022 – December 2024
Research Project — *Python, LLMs, Machine Learning, Code Analysis*
- Developed a 4,500-sample dataset for pairwise code-documentation alignment from 200 open-source Python projects, enabling future research in software maintenance
- Engineered a pipeline for measuring calibration and correctness of large language models for code repair, using Defects4J
- Collaborated in validating efficacy of semantic augmentation of language model prompts for code summarization using precision and recall metrics like ROUGE and METEOR

Teaching Assistant, *University of California, Davis* — September 2023 – December 2023
- Assisted 180 students in a senior-level Probability & Statistical Modeling class.

Publications (Selected)

Toward Reproducible and Standardized Computer Architecture Simulation with gem5, Pai, K., Patel, H., Le, E., et. al., *IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) 2026*
Implications of Full-System Modeling for Superconducting Architectures, Pai, K., Samani, M., Nand, A. & Lowe-Power, J., *Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC Workshops '25)*
CoDocBench: A Dataset for Code-Documentation Alignment in Software Maintenance, Pai, K., Devanbu, P. & Ahmed, T., *International Conference on Mining Software Repositories (MSR) 2025: Data and Tool Showcase Track*
Calibration and Correctness of Language Models for Code, Spiess, C., Gros, D., Pai, K.S., et. al., *International Conference on Software Engineering (ICSE) 2025*
Automatic semantic augmentation of language model prompts (for code summarization), Ahmed, T., Pai, K.S., Devanbu, P. & Barr, E.T., *International Conference on Software Engineering (ICSE) 2024*

Project Experience

NAAMSE: Neural Adversarial Agent Mutation-based Security Evaluator — Nov 2025 – Present
Research Project — *Python, LLMs, Evolutionary Algorithms*

- Won 2nd place (Agent Safety) at the UC Berkeley RDI AgentBeats Competition; framework infrastructure was subsequently forked by Mozilla's 0din team.
- Designed and implemented a clustering engine to identify semantic attack vectors in LLM-generated adversarial prompts
- Engineered an attack pipeline that iteratively generates, evaluates, and refines adversarial prompts against target models
- Benchmarked multiple frontier LLMs within the framework to validate attack effectiveness and refine scoring metrics

HASHIRU: Hierarchical, Resource-Aware Multi-Agent Framework — March 2025 – June 2025
Research Project — *Python, LLMs, Multi-Agent Systems*

- Designed and deployed a multi-agent architecture enabling dynamic, LLM-driven collaboration across diverse tasks
- Implemented task decomposition with intelligent agent delegation based on resource cost models and task specialization
- Engineered autonomous generation of tools and APIs for task execution
- Developed a robust evaluation framework for agent performance across complex, multi-step tasks

SuperNOVA: Superconducting Graph Accelerator in gem5 — January 2024 – March 2026
Research Project, DArchR Lab @ UC Davis — *C++, Python, gem5, Hardware Simulation*

- Modeled a superconducting graph accelerator and interconnect in gem5, achieving up to 24$\times$ speedup and 73$\times$ energy efficiency
- Integrated cryogenic/superconducting cores, caches, and interconnects to evaluate bottlenecks for realistic workloads
- Mentored undergraduates on modeling, benchmarking, and research writing; one co-authored a ModSim 2024 poster

gem5 Vision — January 2023 – June 2023
Resource Discovery Framework — *Next.js, Python, MongoDB, JSON Schema*

- Accelerated resource discovery 20$\times$ for 1,200+ gem5 artifacts by optimizing search and categorization logic
- Implemented categorization and semantic versioning across 20+ resource types to streamline retrieval
- Integrated local and remote JSON schemas with MongoDB, improving accessibility for 500+ users

Awards and Honors

2nd Place (Agent Safety), UC Berkeley RDI AgentBeats Competition — 2026

Dean's List, UC Davis College of Engineering — Fall 2019, Fall 2020, Winter 2022, Spring 2022

Provost Award, UC Davis — 2019–2023

Service

- **Program Committee member**, 23rd International Conference on Mining Software Repositories: Data and Tool Showcase Track
- **Artifact Evaluation Committee member**, 2026 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)
- **Reviewer**, Second Workshop on Agents in the Wild: Safety, Security, and Beyond (ICML 2026)
- **Artifact Evaluation Committee member**, 35th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)